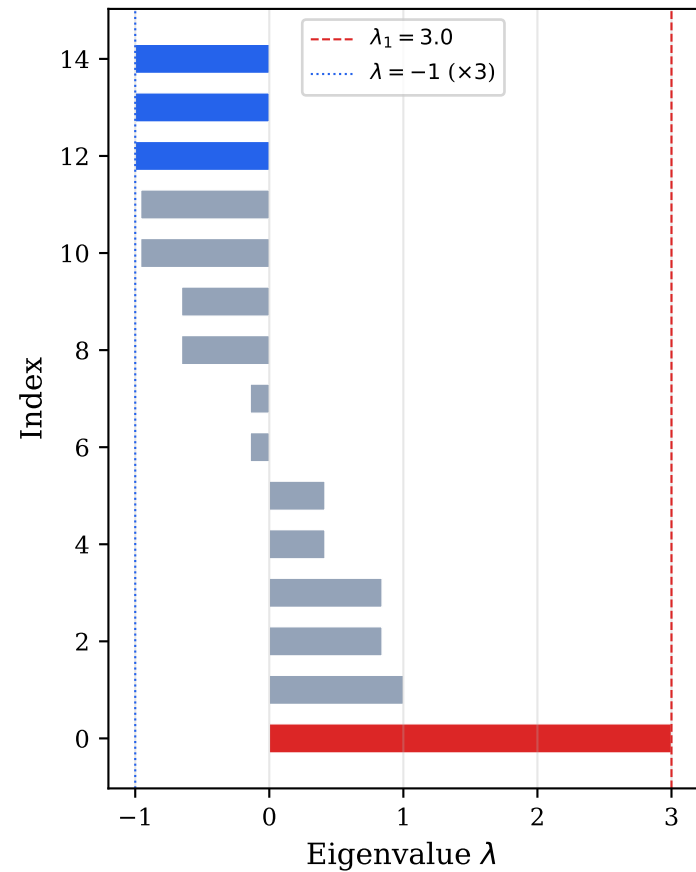
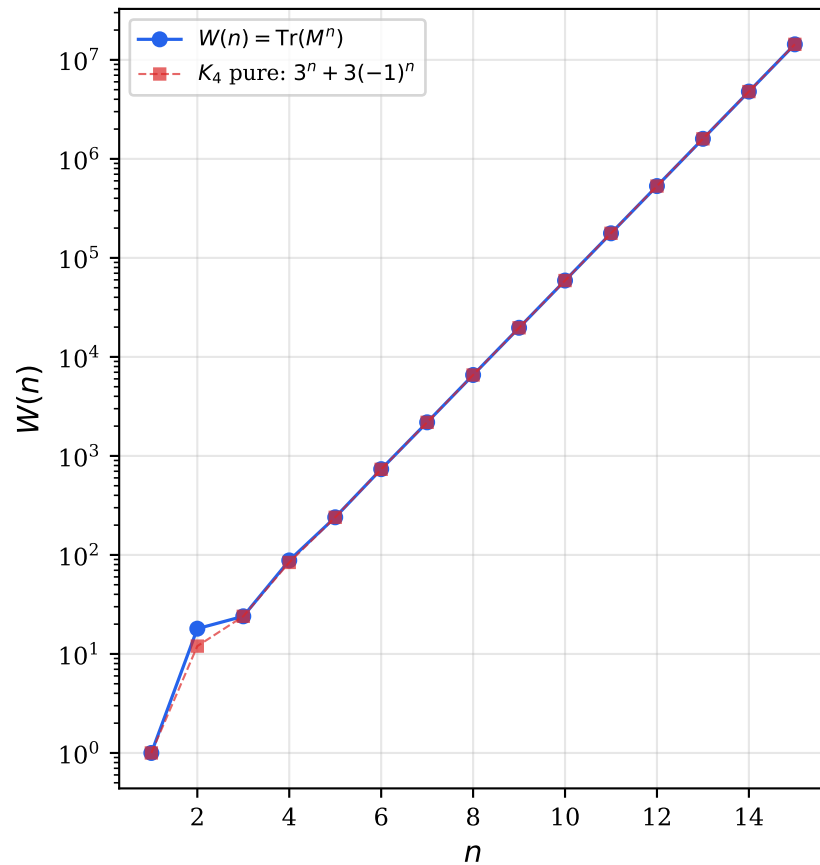


(a) Spectrum of $M_{K_4 + \text{ring}}$ (b) Causal Loop Sequence $W(n)$ 

(c) Vacuum Ring Correction

